

RC-121: Verified Results Summary

Color Sequence

- **Period:** 22 ticks
- **22-tick block:** [1, 2, 1, 3, 0, 0, 1, 3, 1, 2, 1, 4, 3, 1, 2, 1, 4, 2, 1, 3, 0, 0]
- **Color frequencies:** 0: 18.2%, 1: 36.4%, 2: 18.2%, 3: 18.2%, 4: 9.1%

Spectral Analysis

Eigenvalue	Value		
	1.0000	1.0000	0
	0.3173	0.3173	0
	0.0000	0.0000	
	$-0.4086 + 0.1621i$	0.4396	0.8798 5 /6
	$-0.4086 - 0.1621i$	0.4396	-0.8798 -5 /6

- **Stationary distribution:** = [0.182, 0.364, 0.182, 0.182, 0.089]
- **Key ratio:** [1]/ [0] = [1]/ [2] = [1]/ [3] = **2.0** (exact)

Time-Reversal Symmetry

Test	Result
Hamming distance (fwd vs rev)	132/144 = 91.7% different
Transition matrices equal?	No (Frobenius norm = 1.624)
Detailed balance holds?	No (max violation = 0.091)
Entropy production	0.0305 (positive, non-zero)

Falsification Criteria

Criterion	Test	Result
F1	Irreducible (all colors communicate)	PASS
F2	Eigenvalue at 2 /3 (Gram gap)	FAIL (closest: 5 /6)
F3	Non-uniform stationary distribution	PASS
F4	Minimal period = 22	PASS
F5	Time-reversal symmetric	FAIL

Score: 3/5 PASS

Verdict

PARTIAL SUCCESS. The 5-color, period-22 structure is real and structural. The spectral signature is **dodecagonal (5 /6)**, not the Gram gap (2 /3). The dynamics are **irreversible** with positive entropy production — the engine has an intrinsic arrow of time in color space.